

Tag & Track



Conner Museum of Vertebrates

For: Jessica Tir and Katherine Corn

Developers: Martin Hundrup, Shawn Will, and Sarah Carley



Introduction to Tag & Track

- **Objective:** Our objective is to make an iPad app that allows the employees at the Conner Museum to more easily perform their loan tracking.
- **Purpose:** The current way the Museum does things is all on paper, which can get hard to keep track of as well as tedious to work with. Our app aims to make this part of their job less tedious.
- **Scope:** The scope for this project includes loading specimen from arctos, allowing the creating and saving of loans, a way to track what shelf specimen are on, a way to export loan data, as well as a centralized QR code system to make everything quickly scannable.



Team Introduction

- Introductions - Tag and Track Team
 - Martin Hundrup - Team Lead and CS Major, Back End
 - Sarah Carley - CS Major, Front End
 - Shawn Will - CS Major, Database
- Objectives
 - Manual DB entry
 - Signature field for emailer
 - Unit Testing



Sprint Objectives

Defined Objectives

- Manual DB entry
- Signature field for emailer
- Unit Testing

Rationale

- These are core features that needed to be finished so the client had a near complete experience for testing.

Connection to Overall Vision

- These improvements round out the app experience, and allow us to be more confident in the code we have produced.



Feature Implementation and Demo (Martin)

Feature Overview

- Unit Testing (xunit)

Testing and Validation

- 🤔



Feature Implementation and Demo (Martin)

Feature Overview

- Signature Box (handmade)

Testing and Validation

- Functional Testing (UI elements not unit tested)



Feature Implementation and Demo (Shawn)

Feature Overview

- Manual Specimen DB Entry

Testing and Validation

- Manual



Feature Implementation and Demo (Sarah)

Feature Overview

- UI improvements: icons are more used throughout the program, giving it a better look
- Bug fixes: small bug fixes were also performed as they arose to fix certain UI related issues.

Testing and Validation

- Manual testing and validation done by loading up the application.



Deployment of System

- Process of Deployment

- A team member went to the client on Thursday, 3/12 and was successfully able to deploy the application with the client.
- The application was loaded onto the museum's iPad via the development environment.
- During this process, the client was given a small walkthrough of the system so that they could be able to use it easily during testing.

- Impact of Deployment

- The deployment had some issues at first, as the server that was being used for deployment initially was down. Due to this, the deployment had to be delayed to a later date that worked with the client.



Client Feedback on Deployment

Currently, the client has not had enough time with the application to give thorough feedback. We will be collecting in depth feedback all throughout sprint 6 to ensure a great product for the client.



Kanban Overview and Contributions

Kanban Board



MoM Evidence

MoM



Documentation of Report

Report



Responsible Use of AI

Usage

- AI was used at times during development of this project.

Applications and Improvements

- Referenced at times to assist in coming up with approaches to code problems, improving efficiency of certain code by at least 20%.
- Assisted in coming up with the test cases used for the application, finding edge cases quicker and ensuring better test coverage.

Limitations

- We would verify any code made by the AI and do thorough testing to ensure it was sound.

We understand that final decisions and deliverables are our responsibility.



Sprint 5 Achievements and Challenges

What went well

- Manual DB entry for specimen went very well, it only needs a few additions to be complete.
- Unit testing was quick and easy.
- Our commitment to finishing this project kept us going, driving the results we were able to finish.

Challenges Faced

- Signature feature took a few iterations before it worked properly on the iPad.
- Personal events slowed progress.
- Apple server's went down, stopping us from putting our app on the Client's device.



Next Step and Retrospective

Next steps:

- Further improve exporting capabilities.
- Add ability to add specimens manually to loans without the camera.
- Improve the User Interface further with formatting refactors.

Retrospective:

What went well

- Manual DB entry.
- Unit Testing.
- Our perseverance carried us to the finish of this sprint.

What can be Improved

- Improvements to the manual DB entry can be done.
- Various loan logic refining.

Client Feedback:

Client is very happy with the progress and had no extra input. They are very excited to be able to fully use our app.

Sprint 6 Changes

- Updated manual entry.
- CSV export for loans
- Manual additions for loans when camera isn't working



Conclusion

Progress

- Backend and frontend general improvements.
- Signature progress.
- Manual specimen DB entry
- Unit Testing

Key Implementations

- E-mailer now uses the given signature, and has been visually improved.
- Specimen can now be manually added to the DB.

Future Direction

- Exporting loans, specimen information, etc.
- User Interface improvements.

Thank you!